ASSIGNMENT 4

Relational Database vs. NoSQL Database

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons)

Subject: Database Management System

1. Introduction

This report compares relational databases and NoSQL databases and evaluates their suitability for the Hospital Management System. The comparison is technology-neutral; the proposed implementation uses a relational design because its core records have clear relationships and integrity requirements.

2. Comparison

Aspect	Relational database (e.g., MySQL)	NoSQL database (e.g., MongoDB)
Data model	Tables, rows, columns and defined relationships.	Document, key-value, wide-column or graph models.
Schema	Usually explicit schema; changes are controlled through migrations.	Often flexible; documents may have different fields.
Relationships	Foreign keys and joins are native features.	Relationships may be embedded or referenced; joins vary by product.
Integrity	Primary keys, foreign keys, UNIQUE, CHECK and transactions are strong core features.	Capabilities vary by product; some support transactions and validation.
Scaling	Often scales vertically; horizontal scaling is supported by some systems and architectures.	Many products are designed for horizontal distribution, depending on configuration.
Querying	SQL supports joins, grouping, aggregates and subqueries.	Query languages and aggregation APIs depend on the product.
Best fit	Structured records, complex relationships and transactional consistency.	Flexible/high-volume data, rapidly changing structures or particular distributed workloads.

3. Application to hospital data

The core system stores patients, doctors, departments, appointments, prescriptions, bills and feedback. These records depend on each other: an appointment must refer to an existing patient and doctor; a prescription must refer to an existing appointment and medicine; a bill must refer to an appointment. A relational database expresses these dependencies directly with foreign keys and supports multi-table reports with SQL joins.

4. Why organizations choose different technologies

- A hospital billing or appointment system typically prioritizes accurate transactions, consistent identifiers, auditability andwell-defined relationships, favouring a relational database.
- A hospital may use document storage for unstructured clinical notes, device telemetry or flexible forms when fields varysubstantially.
- Large analytics pipelines may use a data warehouse or distributed data platform, while the operational databaseremains relational.
- Selection should be based on consistency needs, data shape, query patterns, scale, operations expertise, cost, securityand recovery requirements—not on the label “SQL” or “NoSQL” alone.

5. Recommendation

Use MySQL 8 as the primary operational database for the eight-table Hospital Management System. It provides relational constraints and SQL joins suited to the schema. NoSQL could complement the system for specific flexible or high-volume workloads, but should not replace the relational core without a concrete requirement and a tested consistency design.

6. Conclusion

Relational and NoSQL systems solve different data-management problems. For this project’s finalized relational schema, a relational database is the most direct fit because correctness of links between appointments, patients, doctors, prescriptions and bills is central.